

# Teacher Guide — primary / module-06- moon-phases-tithi

IKS Curriculum

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## Overview

### Module 6 — Moon Phases & Tithi

**Age band:** Primary · **Grades:** 3–5 · **Duration:** 10 sessions × 30 min (≈ 2 weeks)

### Learning objectives

By the end of this module, children will be able to:

1. **Recognise** the Moon as something they see in the sky, and describe at least four different shapes it takes through a month.
2. **Name** *amāvasyā* (new moon) and *pūrṇimā* (full moon) and say what shape each one is.
3. **Tell** the story: the Sun lights up the Moon, and we see different bits as it travels.
4. **Keep** a 10-night Moon-shape diary, drawing what they see each night.
5. **Say** four of the 15 tithi names — *pratipadā*, *pañcamī*, *aṣṭamī*, *pūrṇimā* — as the “anchor” tithis.

### NEP 2020 alignment

- §4.6 — Integration of Indian Knowledge Systems with early science.
- §4.23 — Experiential learning through nightly Moon observation and torch-and-ball play.
- §4.27 — Local context: children link the Moon to festivals their families celebrate.

### What we don’t teach yet

Reserved for Middle and Senior bands: - The 12° elongation rule - *Kṣaya* and *vṛddhi* tithis - *Yoga*, *karaṇa*, *nakṣatra* (mentioned only in passing if children ask) - *Adhika māsa* - Eclipse geometry (mentioned with safety only) - Cross-cultural calendar comparison

### Prerequisites

- Familiarity with night/day.
- Comfort with drawing simple shapes.
- No Sanskrit knowledge assumed.

## Key vocabulary (one new word per day, repeated)

- *Moon* (English — even Primary children learn the English word fresh too)
- *Sun*
- *amāvasyā* — new moon
- *pūrṇimā* — full moon
- *pakṣa* — fortnight (introduced softly: “the growing half” and “the shrinking half”)
- *tithi* — a “moon day”
- Four anchor tithi names: *pratipadā*, *pañcamī*, *aṣṭamī*, *pūrṇimā*

## Materials checklist (whole module)

- A torch (flashlight)
- A small white ball (a table-tennis ball or styrofoam ball)
- A darkened classroom for 10 minutes
- A printed Moon-shape sticker sheet (8 stickers per child — see Day 5)
- Drawing paper + crayons
- Project supplies (poster paper, glue sticks)

## Day-by-day overview

| Day | Title                                 | Focus                                                               |
|-----|---------------------------------------|---------------------------------------------------------------------|
| 1   | The Moon Story                        | Story + torch-and-ball demo                                         |
| 2   | The Two Halves                        | New moon and full moon; pakṣa idea                                  |
| 3   | The Moon’s House Visits               | The story of the Moon visiting nakṣatra “houses” (folktale framing) |
| 4   | Drawing the Moon                      | Each child sketches 4 phases                                        |
| 5   | The Four Tithi Friends                | Pratipadā, pañcamī, aṣṭamī, pūrṇimā                                 |
| 6   | Festivals and the Moon                | When does Holī happen? When Dīwālī?                                 |
| 7   | Safety + the Moon Goes Dark           | Eclipse safety; “the Sun must never be looked at”                   |
| 8   | Moon Project — Make Your Own Calendar | Sticker calendar                                                    |
| 9   | Moon Project — Decorate and Present   | Posters with phases                                                 |
| 10  | Show and Tell                         | Each child shows their Moon calendar and says one phase name        |

## Project (one line)

Each child makes a “Moon calendar” sticker poster of 10 nights, with the Moon shape and the date for each night. On Day 10 they show it and name two phases.

## A note for teachers

This module is observational and joyful. We don’t compute anything. We don’t test on Sanskrit spelling. The point is that a child leaves this module knowing the Moon changes shape, why (the Sun lights it up), and that there are special names for the new-moon and full-moon nights.

## 10-Day Lesson Plan

### Module 6 (Primary) — 10-Day Lesson Plan

#### At a glance

| Day | Session title               | Lesson file                     | What happens                                          |
|-----|-----------------------------|---------------------------------|-------------------------------------------------------|
| 1   | The Moon Story              | days/day-01-moon-story.md       | Story + torch-and-ball demo                           |
| 2   | The Two Halves              | days/day-02-two-halves.md       | New moon, full moon, growing/shrinking                |
| 3   | The Moon’s House Visits     | days/day-03-house-visits.md     | Folktale framing of the Moon’s monthly journey        |
| 4   | Drawing the Moon            | days/day-04-drawing-the-moon.md | Each child draws 4 phases                             |
| 5   | The Four Tithi Friends      | days/day-05-four-tithis.md      | Naming pratipadā, pañcamī, aṣṭamī, pūrṇimā            |
| 6   | Festivals and the Moon      | days/day-06-festivals.md        | Holī, Dīwālī, Karva Cauth — when do they happen?      |
| 7   | Safety + the Moon Goes Dark | days/day-07-safety.md           | Eclipse safety + brief “sometimes the Moon goes dark” |
| 8   | Make Your Moon Calendar     | days/day-08-make-calendar.md    | Sticker calendar project                              |
| 9   | Decorate and Practise       | days/day-09-decorate.md         | Project finishing + practising “show and tell”        |
| 10  | Show and Tell               | days/day-10-show-and-tell.md    | Each child presents                                   |

#### Pacing notes

- Sessions are 30 minutes, not 45.

- Day 7's safety lesson is the most important non-fun moment. Be firm, then move on.
- Project work fits in Days 8 and 9. Day 10 is presentation only.

## Running theme

Open every class with: **“Did you see the Moon last night? What shape was it?”** Two children share.

## What children leave with

- A 10-night moon diary (with simple sketches)
- A sticker-Moon calendar poster
- Two Sanskrit words they say correctly: *amāvasyā* and *pūrṇimā*
- A torch-and-ball memory of why the Moon changes shape
- The habit of looking up

## Teacher Notes

### Teacher Notes — Module 6 Moon Phases & Tithi (Primary)

#### Posture for this module

Lead with **looking up**. Every class starts with “Did you see the Moon last night?” Children at this age are natural observers. The job here is to give them the language and the explanation, not to test them.

This is also a module where we can be playful — the Moon “visits” its house family, the torch-and-ball demo is just plain fun, festivals come into it. Lean into the playfulness. The Sanskrit terms (only two we insist on — *amāvasyā* and *pūrṇimā*) will stick because they sound interesting and they map to events children's families already celebrate.

#### Common misconceptions to watch for

1. **“The Moon is following me when I walk.”** It's a classic. Don't correct it on Day 1. By Day 4, when children have been observing it across many nights, they'll start to see it's in different places at different times. Let the discovery happen.
2. **“The Moon makes its own light.”** A common belief. The torch-and-ball demo on Day 1 fixes this. If a child says it after Day 1, point at the ball when the torch is off — it goes dark. Reset.
3. **“The Moon disappears at amāvasyā.”** It's still there. We just can't see it because the lit side is facing the Sun and the unlit side faces us. Worth saying explicitly on Day 2.
4. **“Eclipses happen at full moon AND new moon.”** Solar eclipse → new moon (*amāvasyā*). Lunar eclipse → full moon (*pūrṇimā*). At Primary band, we don't go into “why not every month” — we just say “eclipses are rare and special.”

## Pronunciation

Drill these aloud — children will love these two words:

- *amāvasyā* — “ah-MAH-vahs-yah” (4 syllables)
- *pūrṇimā* — “POOR-ni-mah” (3 syllables)

Sing them. Clap them. Make them stick.

## Safety — eclipses

The most important rule of the module:

**NEVER LOOK AT THE SUN. NEVER LOOK AT A SOLAR ECLIPSE.**

Children at this age may have heard their parents prepare for an eclipse. Some families may have observed eclipse rituals. The classroom safety rule is unconditional and takes priority over any other interpretation. Drill it on Day 7. Make every child say it aloud.

Lunar eclipses are safe — children can look at them directly. Make this distinction clearly.

## Religious / secular framing

Several festivals discussed on Day 6 are religious holidays (Holī, Dīwālī, Karva Cauth, Janmāṣṭamī). At Primary band, the framing is **calendrical and cultural**: “These are festivals families in India celebrate, and they all happen on specific Moon-shapes.” We don’t go into the religious meanings unless a child volunteers one — then we acknowledge it (“yes, that’s why some families celebrate it”) and return to the calendar.

If a child’s family does not celebrate any of these festivals, that’s also normal — acknowledge it and ask if they have other festivals on a specific calendar date.

## Differentiation hints

- **Tier up (fast finishers):** Ask them to draw the 8-phase sequence (not just 4). Or, ask them to predict tonight’s Moon shape from the calendar.
- **Tier down (scaffolding):** Reduce the Sanskrit names to just *pūrṇimā* (the easier one). Skip *amāvasyā* if a child is struggling.

## The Moon diary

If clear skies are unavailable (monsoon, light pollution, parent unavailable to take child outside), allow: - Looking it up on an app at home with a parent. - The teacher showing today’s Moon shape from NASA’s calendar at the start of class so children can copy.

The point is the **habit of asking** about the Moon each night. The data accuracy is secondary at Primary band.

## Day 3 — the folktale

Use the story as a teaching device, not as a claim about religious truth. Frame it as “an old story.” Children love it. The astronomical content (the Moon takes ~27 days to come back to the same star) gets remembered better through the story than through a lecture.

## On Sanskrit terms beyond the two

Children may overhear *tithi*, *pakṣa*, *nakṣatra*, etc. from older siblings or parents. If they ask, acknowledge briefly: - *tithi* — “a special Moon-day; we’ll learn more next year.” - *pakṣa* — “the growing half or the shrinking half of the Moon’s month.” - *nakṣatra* — “the Moon’s house in the sky tonight.”

Don’t drill these. *Pūrṇimā* and *amāvasyā* are the two non-negotiables.

## Assessment posture

At Primary band, assessment is sticker-based (see [assessments/rubric.md](#)). No marks, no ranking. Children either showed up and participated, or they need a bit more time.

## Sources

### Sources — Module 6 Moon Phases & Tithi (Primary)

At Primary band, citations are not given to children. Teachers may use the following references for background:

#### For the teacher

- **NCERT Science Class 6**, Chapter 17 (“Stars and the Solar System”) — for the basic Sun–Moon–Earth geometry. Even though Class 6 is the upper boundary of Primary, the diagram and concept of phases are accessible.
- **NASA’s published lunar phase calendars** — search “NASA Moon phases [current year]” — gives a reliable verification of the current month.
- **Any printed *pañcāṅga*** for the current year. Pick one (Drik Panchang, Mypanchang, or a regional almanac) and stick to it for the festivals discussion (Day 6).

#### What we tell children about sources

Nothing. The framing is observational (“look at the Moon tonight”) and story-led (“here is the story of the Moon visiting its house *nakṣatras*”). No citations appear in the children’s workbook or homework.

### The folktale framing (Day 3)

The “Moon visits its house family each night” framing on Day 3 is **a teaching device**, not a folktale we are claiming to source. We tell children that the Indian tradition has 27 *nakṣatras*

(lunar mansions), and many old stories imagine these as 27 houses the Moon visits — one per night.

There ARE traditional stories about the Moon’s relationship with the 27 nakṣatras (in some accounts, the 27 are daughters of Dakṣa, married to the Moon — *Soma* — who favours one over the others, leading to a curse and to the waxing/waning explanation). We tell this **as a story** without claiming a single canonical source — different versions appear across the *Purāṇa* literature. Honest framing: “*There are many old Indian stories about why the Moon changes shape. One of them goes like this...*”

## What we do not source at Primary

We do not cite the *Sūrya-siddhānta* to Primary children. We do not give Sanskrit verse numbers. We do not introduce *adhika māsa*, *yoga*, *karaṇa*, or detailed eclipse mechanics.

## On accuracy

Even though the framing is story-led, every astronomical claim made to children at Primary band is verifiable:

- “The Sun lights up half of the Moon” — true.
- “The Moon takes about 30 days to go around Earth” — close (29.5, which is fine for Primary).
- “There are 15 days between the new moon and the full moon” — close (14.75, fine).
- “Solar eclipses are dangerous to look at” — true.

If a child catches an inaccuracy, that’s wonderful. Acknowledge it and correct gently.

## Day 1 — The Moon Story

**Module:** Moon Phases & Tithi (Primary) · **Time:** 30 min

### Goal

Children meet the Moon as a ball that the Sun lights up, and understand at a basic level why it changes shape.

### Materials

- A torch (flashlight)
- A small white ball (a table-tennis ball or styrofoam ball)
- The room dark enough that the torch beam is visible

### The flow

#### Settle + question (3 min)

Sit on the mat. Teacher: “*Today we are going to meet the Moon.*”

Quick question round: “*Did you see the Moon last night? What shape was it?*” Take 4–5 answers, sketch each one on the board.

### The story (8 min)

#### **The Moon is a ball.**

Imagine a giant white ball, very far away in the sky. That ball is the Moon. It does not make its own light — it is a dark, rocky ball.

Then the Sun shines on it. The Sun is so bright that one half of the Moon lights up. (Hold up the white ball; shine the torch on one side.) Look — half of this ball is bright, and half is dark.

Now — the Moon doesn’t stay still. It travels around the Earth — that’s us. (Walk slowly around a stationary child, holding the ball at arm’s length.)

As the Moon travels, sometimes we see all of the lit side, sometimes only half of the lit side, sometimes none of the lit side. That’s why the Moon looks different every night.

When we see ALL of the lit side, that’s a *pūrṇimā* — a full moon.

When we see NONE of the lit side, that’s an *amāvasyā* — a new moon. The Moon is still there — we just can’t see it.

### The demo (12 min)

Set up: - One child stands in the centre = Earth. - Another child holds the torch = Sun. - The teacher walks the ball around Earth in 4 stopping points: 1. Ball between Sun and Earth → Earth sees the dark side (*amāvasyā*). 2. Ball 90° away → Earth sees half lit (half Moon). 3. Ball opposite the Sun → Earth sees full lit (*pūrṇimā*). 4. Ball 90° away on the other side → Earth sees half lit (other half Moon).

At each stop, ask the Earth-child: “*What do you see?*” The class chorus-answers.

Repeat with a few other children rotating roles. Spend the full 12 minutes — children love this and need the repetition.

### Word practice (4 min)

Chant 3 times together (clap on each syllable):

*am - mah - vāh - yah* — the night with no Moon *poor - ni - mah* — the night with the full Moon

### Wrap-up (3 min)

- Each child draws one Moon shape on paper — whichever shape they remember best from the demo.
- Hold up 3 drawings.
- **Homework:** *Look at the Moon tonight. Whatever shape you see, draw it in your workbook. If it’s cloudy, ask a grown-up to look it up on a phone.*



## Homework / take-home

- **Start your Moon Diary tonight.** Look up. Draw what you see. Note: clear or cloudy.

## Differentiation

- **Tier up:** Draw two phases (last night's + tonight's predicted).
- **Tier down:** Just remember the word *pūrṇimā* (the easier one). *Amāvasyā* can come later this week.

## Teacher's quick notes

- The single most important sentence today: **“The Sun lights the Moon. The Moon doesn't make its own light.”** Repeat it often.
- Some children will need the demo repeated. That's fine — that's the whole class.
- Don't worry about Sanskrit spelling. They'll get it through repetition over the week.

## Day 2 — The Two Halves

**Module:** Moon Phases & Tithi (Primary) · **Time:** 30 min

### Goal

Children learn that the month has two halves — one where the Moon grows, one where it shrinks — and that *amāvasyā* and *pūrṇimā* are the special anchor nights.

### Materials

- The torch and ball from Day 1
- A long strip of paper or a chalk line on the floor
- Drawing paper + crayons

### The flow

#### Settle + recall (4 min)

Sit on the mat. Teacher: “*Did anyone see the Moon last night?*” Take 3–4 answers. “*Did you draw it in your diary?*” Show one or two diary drawings.

Chant once together: “*am - mah - vahs - yah! poor - ni - mah!*”

#### Story (6 min)

**The Moon takes a journey.**

The Moon goes around the Earth — but it takes a long time. About 30 days.

Halfway through the journey, the Moon is on the OPPOSITE side from the Sun. That's when we see the WHOLE bright side. That's *pūrṇimā*.

At the start of the journey — and again at the end — the Moon is on the SAME side as the Sun. We can't see the bright side at all. That's *amāvasyā*.

So between *amāvasyā* and *pūrṇimā* — the FIRST half of the journey — the Moon gets bigger and bigger every night. That’s the **growing half**.

Between *pūrṇimā* and *amāvasyā* — the SECOND half of the journey — the Moon gets smaller and smaller. That’s the **shrinking half**.

### Demo + walk (10 min)

Draw a long line on the floor (or on the board). At the left end write *amāvasyā* and draw a dark circle. At the right end write *pūrṇimā* and draw a bright full circle.

In the middle, draw 5 stages between them — a thin sliver, a half-moon, a 3/4 moon, etc.

Now do the same for the second half — *pūrṇimā* shrinks back through 3/4, half, sliver, to *amāvasyā*.

Have 5 children walk along the line, holding a paper Moon shape at each station. The rest of the class chants “GROWING” while they walk left-to-right and “SHRINKING” while they walk right-to-left.

### Drawing (7 min)

Each child draws TWO Moons on paper: 1. A *pūrṇimā* — a full bright circle. Label it “*pūrṇimā*.” 2. An *amāvasyā* — an empty circle (just the outline, dark inside, or a black filled circle with a tiny sliver). Label it “*amāvasyā*.”

### Wrap-up (3 min)

- Hold up 2–3 drawings.
- Chant once more: “*am - mah - vahs - yah! poor - ni - mah!*”
- **Homework:** Look at the Moon tonight. Is the Moon growing or shrinking? (Ask a grown-up if you don’t know.)

### Homework

- Continue the Moon Diary. Add “**growing**” or “**shrinking**” next to tonight’s sketch.

### Differentiation

- **Tier up:** Draw all the Moon shapes on the line — not just *pūrṇimā* and *amāvasyā*.
- **Tier down:** Just draw *pūrṇimā* today. *Amāvasyā* tomorrow.

### Teacher’s quick notes

- Some children will say “the Moon is hiding” on *amāvasyā*. That’s a lovely phrasing — agree, then add: “and the Sun’s bright light is making it hard for us to see anyway.”
- Watch for the misconception “the Earth’s shadow makes the new moon.” It doesn’t (that’s eclipses). Today we don’t unpack that — just say “no, it’s because we can’t see the bright side.”
- The words “growing” and “shrinking” are easier than *śukla pakṣa* / *kṛṣṇa pakṣa*. Stick with English at Primary.

# Day 3 — The Moon’s House Visits

**Module:** Moon Phases & Tithi (Primary) · **Time:** 30 min

## Goal

Children meet the idea that the Moon visits a different “house” (star pattern) every night — a folktale-style framing of the 27 *nakṣatras*. The Sanskrit word *nakṣatra* is introduced softly without drilling.

## Materials

- A printed Moon and 27 paper “house” stars arranged in a circle on the floor (or on a wall poster)
- Drawing paper

## The flow

### Settle + recall (3 min)

Chant: “*am - mah - vāh - yah! poor - ni - mah!*” — three times.

Recall: “*Is the Moon growing or shrinking right now?*” (Refer to the day’s actual sky if children have observed.)

### Story (12 min)

#### **The Moon Visits Its House Family**

Long, long ago, the Moon — whose name is *Soma* — had 27 sisters. Each sister lived in her own house, high up in the sky. Together they were the family of *nakṣatras* — the “star houses.”

Every night, the Moon visited ONE of his 27 sister houses. He never visited two on the same night. So one night he was in the house called *Aśvinī*. The next night, *Bharāṇī*. The next, *Kṛttikā*. And so on, all 27 sisters, one per night.

After 27 nights, he had visited every sister. Then he started over — back to *Aśvinī*.

*(Children love this. Walk a paper Moon around the 27 stars on the floor, naming a few of them as you go.)*

Now — the Moon also changes shape every night, as we learned. Sometimes he is full (*pūrṇimā*), sometimes he disappears (*amāvasyā*). Some old stories say his sisters are happy when he is full, and worried when he disappears.

But every month, he comes back to the same sister he visited a month ago. So if tonight the Moon is in the house of *Citrā*, in about a month, the Moon will visit *Citrā* again.

That’s why our months in India are named after the sisters. The month *Caitra* is named after *Citrā*. The month *Kārtika* is named after *Kṛttikā*. The Moon’s sisters give the months their names.

## Activity (10 min)

Each child: 1. Draws the Moon in the middle of a paper. 2. Draws 4 little “house stars” around the Moon (we’ll do 4, not 27). 3. Labels one of the houses with the name *Aśvinī* (or any nakṣatra name the teacher chooses). 4. Draws arrows showing the Moon’s path from one house to the next.

This is a drawing exercise. Don’t drill the names.

## Wrap-up (5 min)

- Show a few drawings.
- “What did we learn? The Moon visits a new house every night. There are 27 houses in all. The Moon’s sisters give our months their names.”
- **Homework:** Look at the Moon tonight. What shape? Write the date.

## Homework

- Continue the Moon Diary.

## Differentiation

- **Tier up:** Try drawing all 27 houses in a ring around the Moon.
- **Tier down:** Just draw the Moon and one house. The story is the lesson — the drawing supports it.

## Teacher’s quick notes

- The “27 sisters” framing is a teaching tool drawing on traditional stories (the Dakṣa’s daughters narrative appears in several *Purāṇas*, with the Moon as *Soma* / Candra). At Primary band, frame it as “an old story” — don’t claim a single canonical source.
- The point: children learn that the Moon’s position in the sky changes night by night, in a 27-night cycle (the *sidereal* month — different from the 30-day *synodic* month, but we don’t need that distinction yet).
- If a child asks “why don’t the houses come back to the same place every 30 nights?” — that’s a great question, and you can tell them they’ll learn about it in Middle school. (The answer: because Earth has moved around the Sun during the 27 days too.)
- Some children may know specific nakṣatras from their birth chart. That’s fine — let them share, but don’t dwell on astrology.

# Day 4 — Building a Tithi Calendar

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

## Goal

Construct a tithi calendar for the current month using observation and computation.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: "What's one thing you remember from yesterday?"

### Today's idea (10 min)

- *kṛṣṇa pakṣa* (IAST: kṛṣṇa pakṣa; lit. "waning fortnight (full → new)")
- Let's make a moon calendar for this month!

Show a printed month calendar. Together, look up today's tithi. Mark it with a moon-sticker.

Each child gets a worksheet with 28 boxes (one per day). They draw the moon shape they expect to see each evening, going from now to a month from now. Tomorrow we'll check and correct.

### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See [homework.md](#).)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 5 — Lunar Months and Festivals

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

## Goal

Map the 12 lunar months (caitra, vaiśākha, ...) to major festivals.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

### Today's idea (10 min)

- *tithi* (IAST: tithi; lit. “lunar day (1/30 of synodic month)”)
- The year has 12 moon-months, each with its own name and feeling:
  - *Caitra* (spring)
  - *Śrāvaṇa* (monsoon)
  - *Kārtika* (Diwali month)
  - *Phālguna* (Holi month) — and many more!

Show pictures of 4 festivals. Ask: “*Which moon-month is this festival in?*” Children sort cards.

Sing the months: “*Caitra, Vaiśākha, Jyaiṣṭha, Āṣāḍha...*” (Through all 12.)

### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See [homework.md](#).)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

# Day 6 — Panchanga — Five Limbs

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

## Goal

Identify the five components: tithi, vāra, nakṣatra, yoga, karaṇa.

## Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

## The flow

### Settle + chant (3 min)

Daily chant or breathing exercise.

### Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

### Today's idea (10 min)

- *amāvasyā* (IAST: amāvasyā; lit. “new moon”)
- The pañcāṅga has 5 parts — like 5 fingers on a hand!
  1. *tithi* — moon-day
  2. *vāra* — weekday (Mon, Tue, ...)
  3. *nakṣatra* — star-house the moon is visiting
  4. *yoga* — Sun-Moon meeting
  5. *karaṇa* — half-moon-day

Show today's pañcāṅga on the board. Children copy the 5 names into their workbook with one word each.

### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

## Homework

(See [homework.md](#).)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

## Day 7 — Why Eclipses Happen

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

### Goal

Explain solar/lunar eclipses using orbital plane geometry (5° tilt of Moon's orbit).

### Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

### The flow

#### Settle + chant (3 min)

Daily chant or breathing exercise.

#### Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

#### Today's idea (10 min)

- *pūrṇimā* (IAST: pūrṇimā; lit. “full moon”)
- Sometimes the Moon hides the Sun — that's a *solar eclipse*!
- Sometimes the Earth's shadow falls on the Moon — that's a *lunar eclipse*!

Use the torch + ball + child's head from earlier days. Show: - Ball blocks light → solar eclipse (child's face goes dark). - Child's head blocks light → lunar eclipse (ball goes dark).

**SAFETY RULE:** Never look at the Sun with eyes! Children repeat 3 times.

#### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

#### Wrap-up (3 min)

Each child shares one word that describes today.



## Homework

(See homework.md.)

## Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

## Day 8 — Modern Lunar Science

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

### Goal

Phases as Sun's illumination of Moon; mention libration, tides.

### Materials

- Notebook, pencils, drawing paper
- (See lesson-plan.md for any day-specific materials)

### The flow

#### Settle + chant (3 min)

Daily chant or breathing exercise.

#### Story recall (4 min)

Briefly recall yesterday's lesson by asking: “*What's one thing you remember from yesterday?*”

#### Today's idea (10 min)

- *pañcāṅga* (IAST: pañcāṅga; lit. “five-limbed Indian calendar (tithi, vāra, nakṣatra, yoga, karaṇa)”)
- The Moon isn't really changing shape — the same ball is always there!
- The Sun lights up half of it. We see *different parts of the lit half* as the Moon goes around Earth.

Demo: hold a half-painted ball (one side bright, one side dark). Walk it around a child. The child describes what they see from each angle.

“*The Moon is always whole. We only see the bright part.*”

### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

### Homework

(See [homework.md](#).)

### Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

## Day 9 — Cross-cultural Calendars

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

### Goal

Compare with Hijri (Islamic), Chinese, and Hebrew lunar calendars.

### Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

### The flow

#### Settle + chant (3 min)

Daily chant or breathing exercise.

#### Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

#### Today’s idea (10 min)

- *candra* (IAST: candra; lit. “Moon”)
- People all over the world watch the Moon, but each tradition counts months a bit differently.
- Some use only Moon (Islamic Hijri). Some use Sun + Moon (Indian, Chinese, Hebrew). Some use only Sun (modern Gregorian — like the school year).

Show 4 calendar pictures and chant: “*India and China — Moon plus Sun! Islam — Moon alone! Modern world — Sun alone!*”

Each child colours a small “moon flag” with the calendar they think they use most.

### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

### Homework

(See [homework.md](#).)

### Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

## Day 10 — Synthesis and Quiz

**Module:** Moon Phases and Tithi (Primary) · **Time:** 30 min

### Goal

Final quiz and 10-day moon-diary share.

### Materials

- Notebook, pencils, drawing paper
- (See [lesson-plan.md](#) for any day-specific materials)

### The flow

#### Settle + chant (3 min)

Daily chant or breathing exercise.

#### Story recall (4 min)

Briefly recall yesterday’s lesson by asking: “*What’s one thing you remember from yesterday?*”

#### Today’s idea (10 min)

- *pakṣa* (IAST: pakṣa; lit. “fortnight (half-month)”)

- Today we celebrate everything we learned about the Moon!

Each child shares ONE thing from their moon diary. “*Did the Moon look how you thought it would?*”

Make a class moon-mural: a circle with all the children’s moon drawings stuck around the outside. Sing the tithi rhyme one more time.

Take the quiz with stickers (not stress) — every child gets a “Moon Watcher” sticker.

### Activity (8 min)

A simplified version of *10-Day Moon Observation Diary* (see [activities/](#)). Draw, sing, or act it out.

### Wrap-up (3 min)

Each child shares one word that describes today.

### Homework

(See [homework.md](#).)

### Teacher notes

- Citations belong in teacher notes only, not in student-facing material at this band.
- Keep new Sanskrit terms to one per lesson, repeated several times.
- If energy is low, swap the activity for free drawing.

## Formative Quiz — Moon Phases and Tithi (Primary)

**When:** mid-module (after Day 5). **Duration:** 20 minutes. **Format:** mix of short answer, multiple choice, and reflection. Sticker self-assessment for Primary.

### Questions

1. Name the three / four key terms introduced in this module so far. Give the IAST and one-line meaning of each.
2. State the central idea of Moon Phases and Tithi in one sentence.
3. (Multiple choice) Which day in this module focused on ‘Fifteen Tithis per Pakṣa’? (a) Day 1 (b) Day 2 (c) Day 3 (d) Day 4
4. True / False — and explain in one sentence: *Moon Phases and Tithi is purely a religious concept and has no scientific framing.*
5. Connect Moon Phases and Tithi to **one** other module in this curriculum. Which module, and how?
6. List **two** sources cited so far in this module.
7. (Short answer) Why does the module distinguish ‘candra’ from related concepts? Give one example.

8. (Short answer) In your own words, explain Day 3's concept.
9. Identify **one** common misconception about Moon Phases and Tithi and rebut it in one sentence.
10. Write **one** question you have about this module so far — something you want clarified before the summative.

## How this is used

This quiz is **formative** — it's a check-in to see what's landing, not a grade. Teacher reviews answers to adjust pacing for Days 6–10. Students may swap and self-mark.

## Summative Quiz — Moon Phases and Tithi (Primary)

**When:** end of module (after Day 10). **Duration:** 30 minutes. **Total points:** 20.

### Questions

**Part A — Vocabulary (~5 points)** A1. Define *candra* in your own words. A2. Define *pakṣa* in your own words. A3. Define *śukla pakṣa* in your own words. A4. Define *kṛṣṇa pakṣa* in your own words. A5. Define *tithi* in your own words.

**Part B — Concepts (~10 points)** B1. Explain the central idea of Moon Phases and Tithi in your own words (3–4 sentences). B2. Describe the method or framework students used in Day 4 of this module. B3. Identify ONE source cited in this module and explain why it matters. B4. Compare Moon Phases and Tithi with a concept from another module (your choice). What's similar, what's different? B5. Identify ONE common misconception about this module and rebut it in 2–3 sentences.

**Part C — Application (~5 points)** C1. Give an example of Moon Phases and Tithi from your own daily life. C2. Describe one way this module connects to a current real-world issue (or scientific question).

**Part D — Reflection (~5 points; ungraded for content)** D1. What surprised you most in this module? D2. What's one question you still have?

### Marking guidance

- Vocabulary: give credit for correct meaning, even if exact wording differs.
- Concepts: look for evidence of understanding, not memorisation.
- Application: any reasonable example accepted.
- Reflection: ungraded for content; awarded for thoughtfulness.

### Pass mark

A score of 12/20 indicates the student has met the module's learning objectives.

# Homework

## Homework — Moon Phases and Tithi (Primary)

Light daily tasks. Each takes 5–10 minutes.

1. **Day 1 — The Moon and Its Shapes** · Draw about today's concept. (~5 min)
2. **Day 2 — Two Fortnights — Pakṣa** · Draw about today's concept. (~5 min)
3. **Day 3 — Fifteen Tithis per Pakṣa** · Draw about today's concept. (~5 min)
4. **Day 4 — Building a Tithi Calendar** · Draw about today's concept. (~5 min)
5. **Day 5 — Lunar Months and Festivals** · Draw about today's concept. (~5 min)
6. **Day 6 — Panchanga — Five Limbs** · Draw about today's concept. (~5 min)
7. **Day 7 — Why Eclipses Happen** · Draw about today's concept. (~5 min)
8. **Day 8 — Modern Lunar Science** · Draw about today's concept. (~5 min)
9. **Day 9 — Cross-cultural Calendars** · Draw about today's concept. (~5 min)
10. **Day 10 — Synthesis and Quiz** · Draw about today's concept. (~5 min)

## Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.

## Activity 1 — 10-Day Moon Observation Diary

**Module:** Moon Phases and Tithi · **Band:** Primary · **Duration:** 30 min

### Purpose

Reinforce the day-by-day concepts of Moon Phases and Tithi through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

### Materials

notebook, pencils, ~15 minutes per evening

### What students do

Each evening for 10 days, students sketch the moon shape they see, note the time, the tithi name, and one feeling/observation. On Day 10 they compare sketches with the actual calendar.

### Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.

2. **Working phase (20 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

## Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

## Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

## Safety

- General safety briefing before any materials are handled.
- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

## Project Brief

### Project Brief — Class Tithi Calendar for the Current Month

**Module:** Moon Phases and Tithi · **Band:** Primary

#### What you will do

Each student picks 3 dates in the current calendar month. They compute (or look up) the tithi, identify the pakṣa, and produce a visual class calendar with all entries combined.

#### Why

This project asks you to apply concepts from this module to your own life or a real situation. It is graded on the rubric in `rubric.md`.

#### Deliverables

- **One-page project plan** (Day 2 of the module): problem, sources, deliverable, timeline.
- **Daily journal entries** (one per day, starting Day 6): what you did, what worked, what didn't.
- **Final artefact** (a drawing / made object).
- **Written reflection** (oral share): what you learned, what surprised you, what you're still uncertain about.
- **Presentation** (3-minute share with the class).

## Timeline

- **Day 1–2:** scope and plan.
- **Day 3–4:** research.
- **Day 5:** mid-project critique.
- **Day 6–7:** build / make.
- **Day 8:** document.
- **Day 9:** rehearse.
- **Day 10:** share day.

## Sources

You must cite at least 1 sources. Use the form: *Author, Title, year*. (See `../sources.md` for examples.)

## Safety

- Any project involving plants / food / outdoor work must be supervised.
- **No herbal medicine recommendations.** Modules 10 and 4 give awareness; therapeutic decisions belong to qualified practitioners.

## Grading

See `rubric.md`. The project is worth the same weight as the summative quiz.

## Rubric

### Assessment Rubric — Moon Phases and Tithi (Primary)

This rubric applies to the project (see `project-brief.md`) and may also inform the summative quiz.

#### Criteria

- **Knows the new word(s)** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Joined in activities** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Asked good questions** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice
- **Drew / made something** — 🌟 Strong understanding / 🌱 Growing / ☁️ Needs more practice

#### How the teacher uses this rubric

1. After the project share, score each student on each criterion.
2. Convert to a band (4 = Exceeding, 3 = Meeting, etc.) for the report card.



3. Share at least one strength and one area for growth with each student.

### Self-assessment (Senior students)

Senior students complete a parallel self-assessment using the same rubric before the teacher's assessment. The teacher and student then meet to compare.